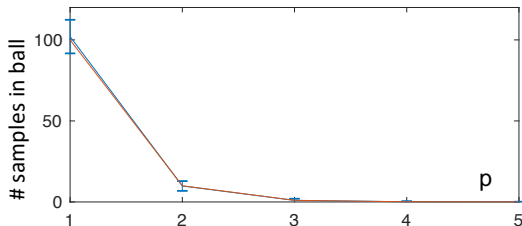


Section 2D. Parametric Models

Victor M. Preciado, PhD
Dept of Electrical & Systems Engineering
University of Pennsylvania
preciado@seas.upenn.edu

Curse of Dimensionality: Recap

CoD: When the number of inputs p in your problem is large, the CoD does not allow you to use local averaging around a point $\mathbf{x}$ to estimate the regression function $f(\mathbf{x})$



- How can we beat the CoD? Parametric models...

Parametric Models

To overcome the curse of dimensionality, we use parametric models. For example, *linear models* of the form:

$$f_L(\mathbf{x}; \beta) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p$$

Reminder. $\mathbf{x} = (x_1, \dots, x_p)^\top$; hence x_i are not samples, but the components of the vector $\mathbf{x}$

- ▶ We can estimate the parameters β_i from a dataset of observations $\mathcal{D}$. We will denote the estimated parameters as $\hat{\beta}_i$ and the estimated linear function as

$$\hat{f}_L(\mathbf{x}; \beta) = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \cdots + \hat{\beta}_p x_p$$

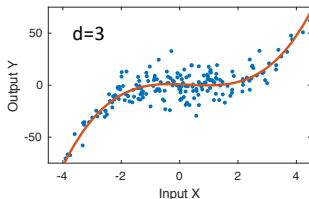
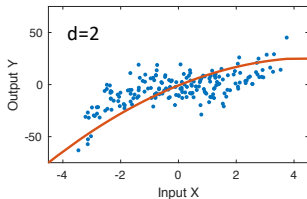
- ▶ The main reason why this model does not suffer from CoD is that to estimate the parameters β_i and to make predictions we make use of the whole dataset $\mathcal{D}$, instead of a small number of samples.

Parametric Models (cont.)

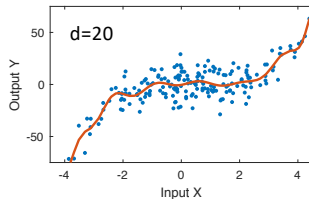
- ▶ During this course, we will see a variety of parametric models: Linear models, polynomial models, regression trees, support vector machines, etc.
- ▶ For example, for $p = 1$, a polynomial model takes the form

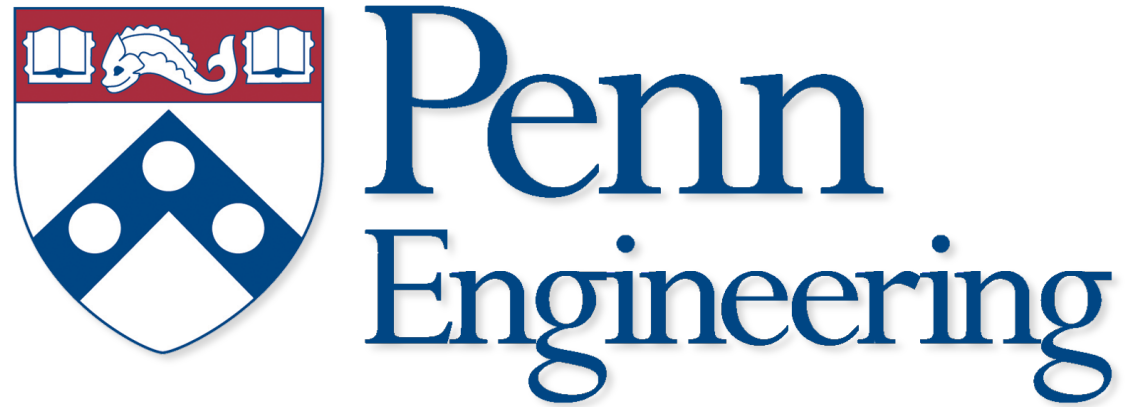
$$\hat{f}_L(X; \alpha) = \hat{\alpha}_0 + \hat{\alpha}_1 X + \hat{\alpha}_2 X^2 + \dots + \hat{\alpha}_d X^d$$

where d is the degree of the polynomial model. The larger the d , the more *flexible* our model is, since it can learn functions with more and more oscillations.



...





Copyright 2020 University of Pennsylvania
No reproduction or distribution without permission.